

IS 107 – Business Intelligence

Laboratory Work 4

Forecasting Time-Series Data in Power BI

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Course: IS 107 – Business Intelligence

Forecasting

Time-Series

Power BI

Analytics

Lab 4 Overview

Part 1: Tourism Data

- Download tourism-data.csv
- Import into Power BI
- Create Line Chart (Yearly)
- Add Trend Line
- Add Forecast (12 pts, 95% CI, Seasonality 12)
- Analyze forecast for 2020

Part 2: Imperfect Data

- Use tourism-data-modified.csv
- Dataset includes COVID disruption (2020–2021)
- 50 new data points added
- Experiment with forecast settings
- Analyze confidence intervals
- Compare Ignore Last settings

Analysis & Report

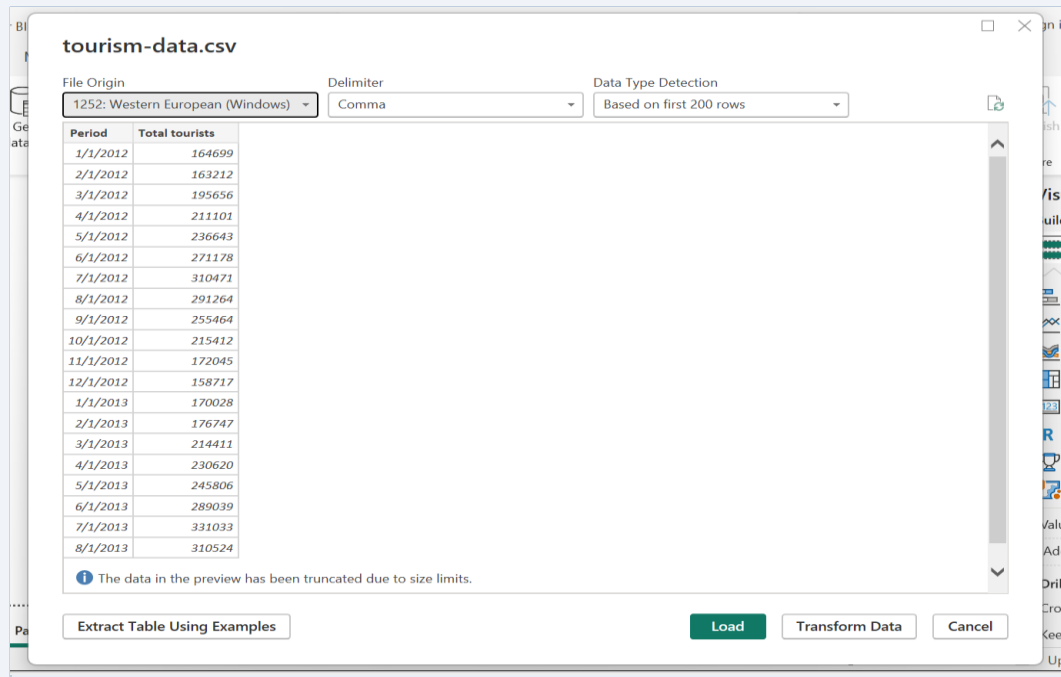
- Describe dataset & source
- Visualize forecast with screenshots
- Discuss trend & seasonality
- Interpret confidence intervals
- Discuss disruption handling
- Conclusions & limitations

PART 1

Tourism Data Forecast

Dataset: tourism-data.csv • Objective: Apply Forecasting in Power BI

Step 1–2: Import Data & Create Line Chart



Step 1 – Download Dataset

Go to the GitHub link and save tourism-data.csv to your computer.

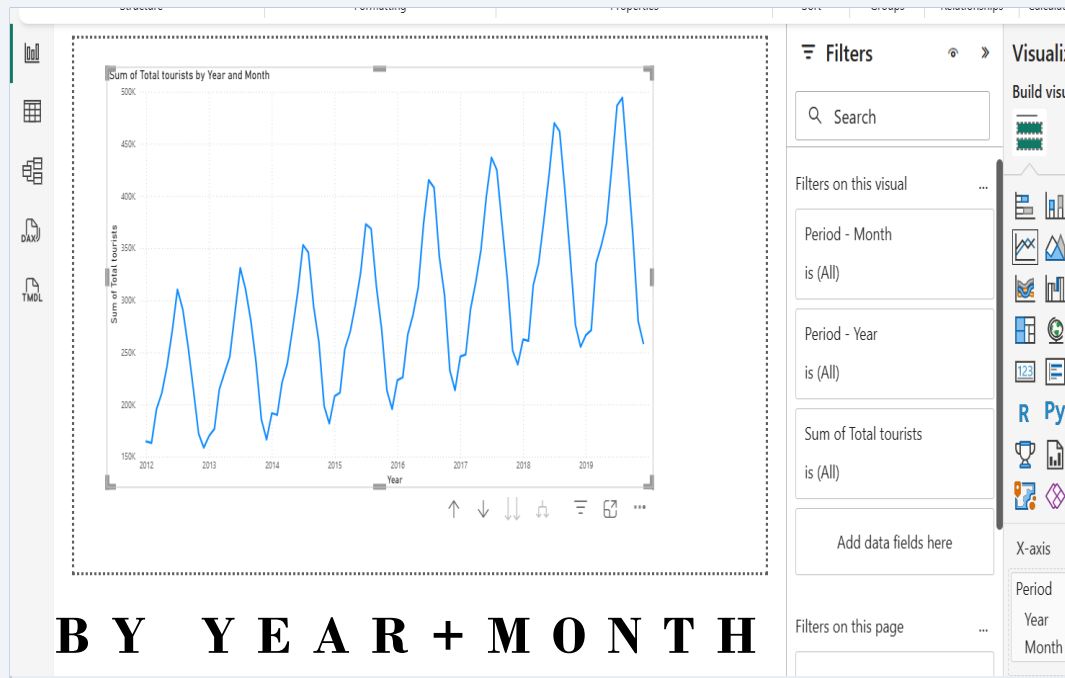
Step 2 – Import into Power BI

Home → Get Data → Text/CSV → Load tourism-data.csv

Step 3 – Create Line Chart

- Drag Period → X-axis
- Drag Total tourists → Y-axis
- Set hierarchy to Year + Month

Step 1–3: Import Data & Create Line Chart



Step 1 – Download Dataset

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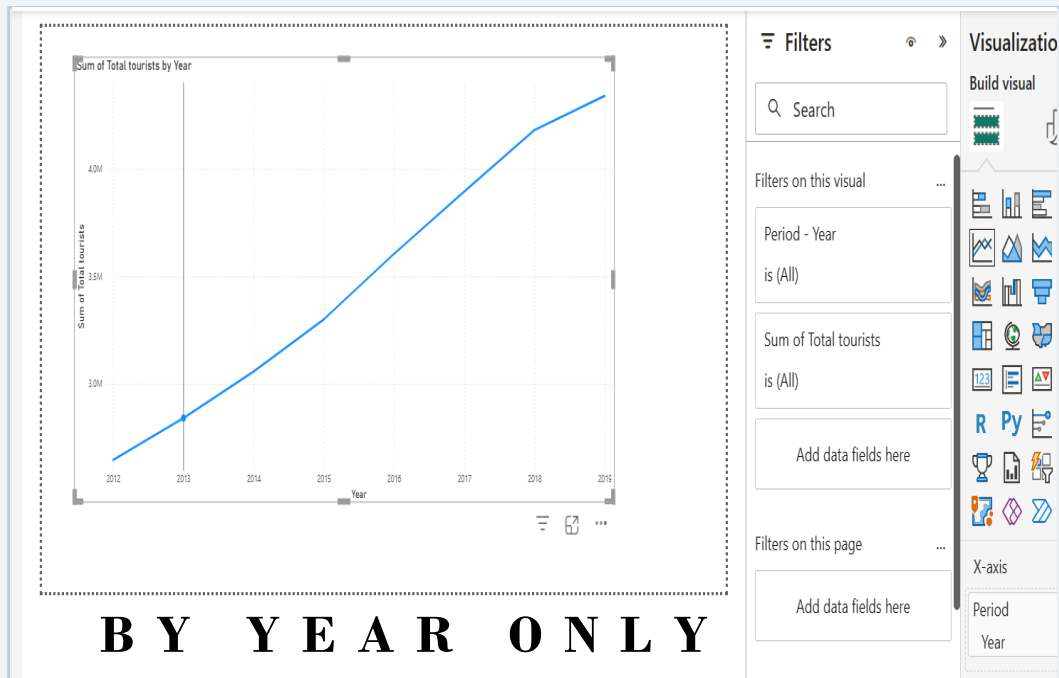
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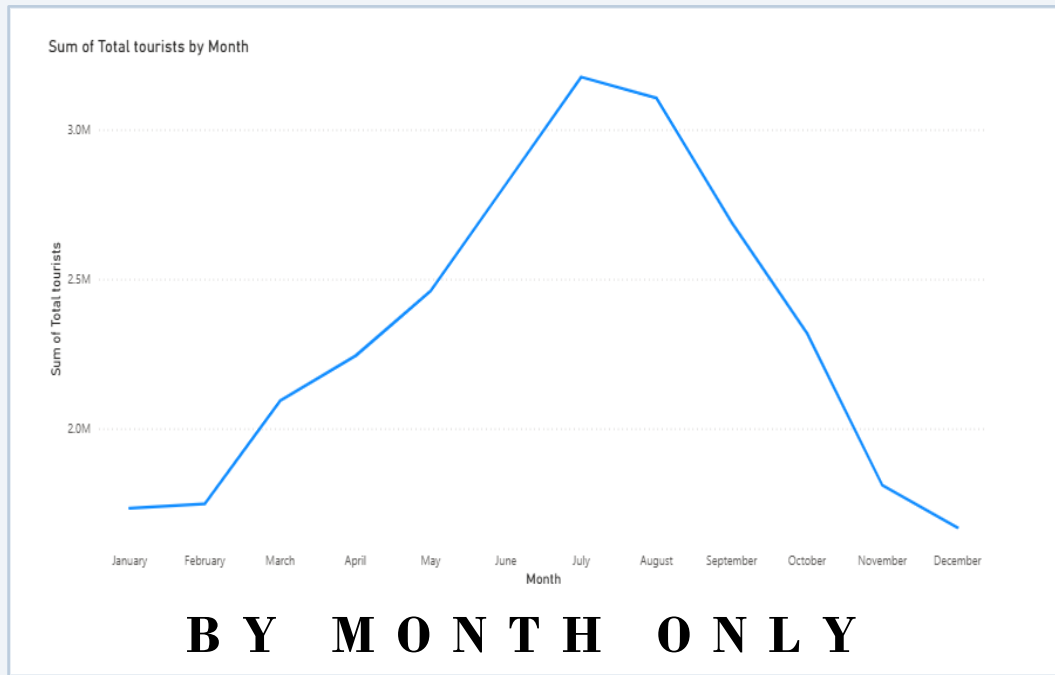
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Step 1–3: Import Data & Create Line Chart



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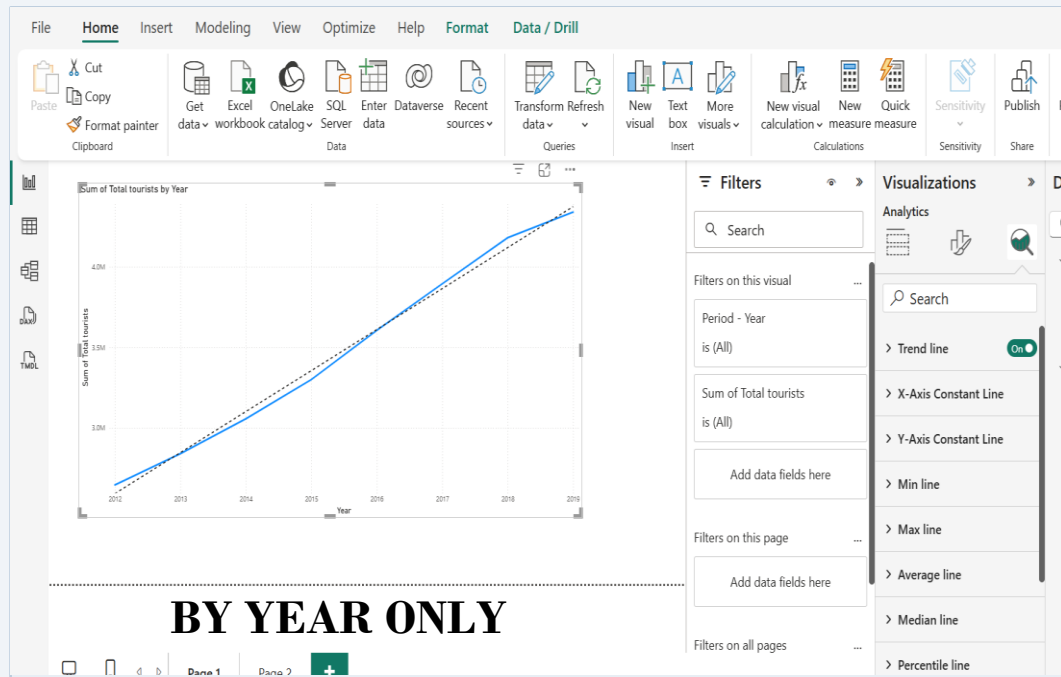
Step 2 – Import into Power BI

Home → Get Data → Text/CSV → Load tourism-data.csv

Step 3 – Create Line Chart

- Drag Period → X-axis
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- Set hierarchy to Year + Month

Step 4: Add a Trend Line



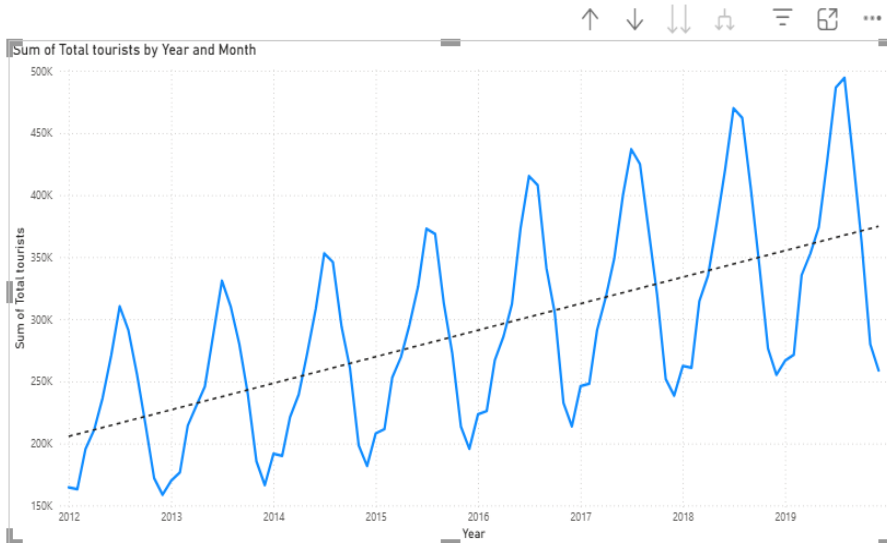
How to Add Trend Line

1. Click the line chart to select it
2. Click the Analytics icon () in Visualizations pane
3. Expand 'Trend line'
4. Click + Add

What You Should See

A dashed diagonal line appears across the chart, confirming the strong upward trend in total tourism from 2012 to 2019.

Step 4: Add a Trend Line



BY MONTH + YEAR

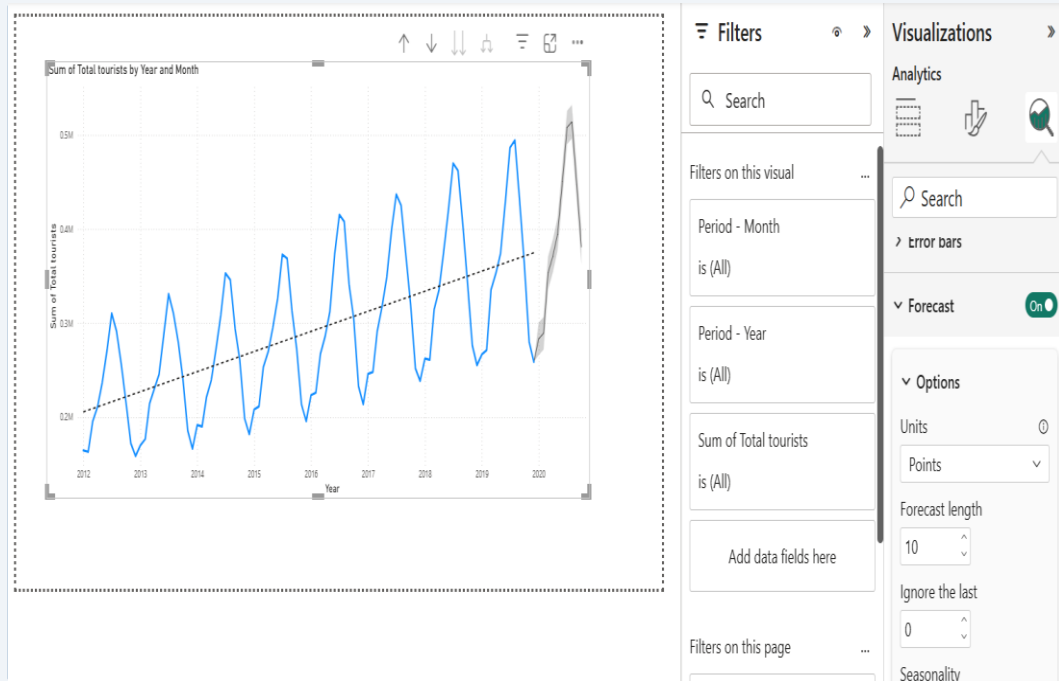
How to Add Trend Line

1. Click the line chart to select it
2. Click the Analytics icon () in Visualizations pane
3. Expand 'Trend line'
4. Click + Add

What You Should See

A different line appears across the chart, confirming the strong upward trend in total tourism from 2012 to 2019.

Step 5: Add Forecasting



Forecast Configuration Settings

Forecast length **10 points**

Ignore last **0 points**

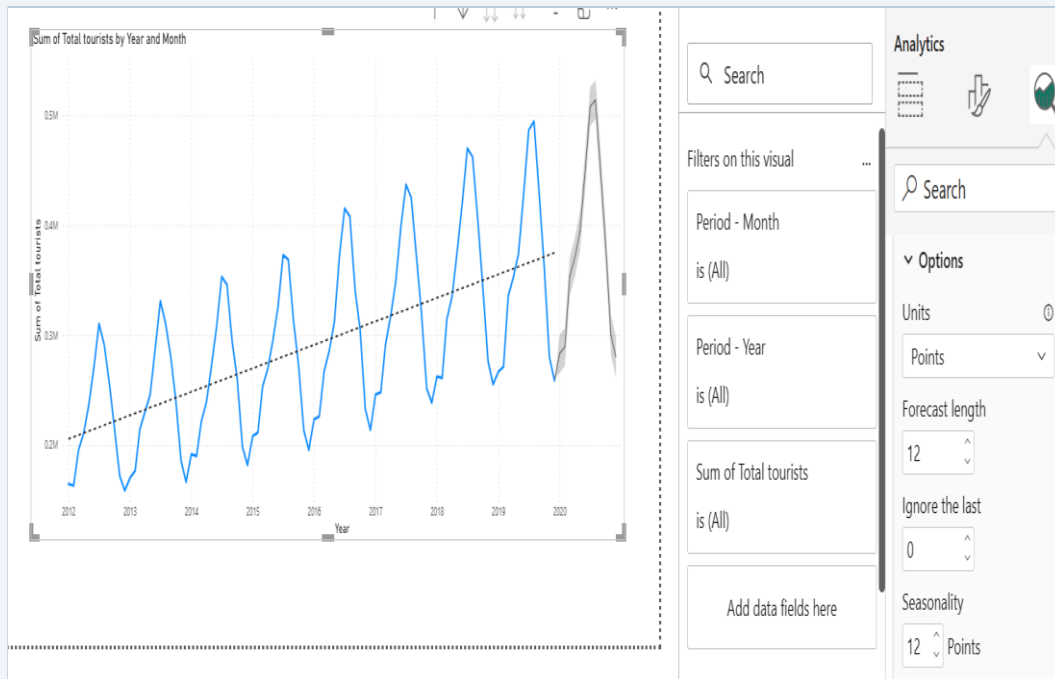
Confidence interval **95%**

Seasonality **..**

Then Click Apply

The forecast extends beyond 2019 as a gray shaded area representing the 95% confidence interval.

Step 6: Configure the Forecast



Forecast Configuration Settings

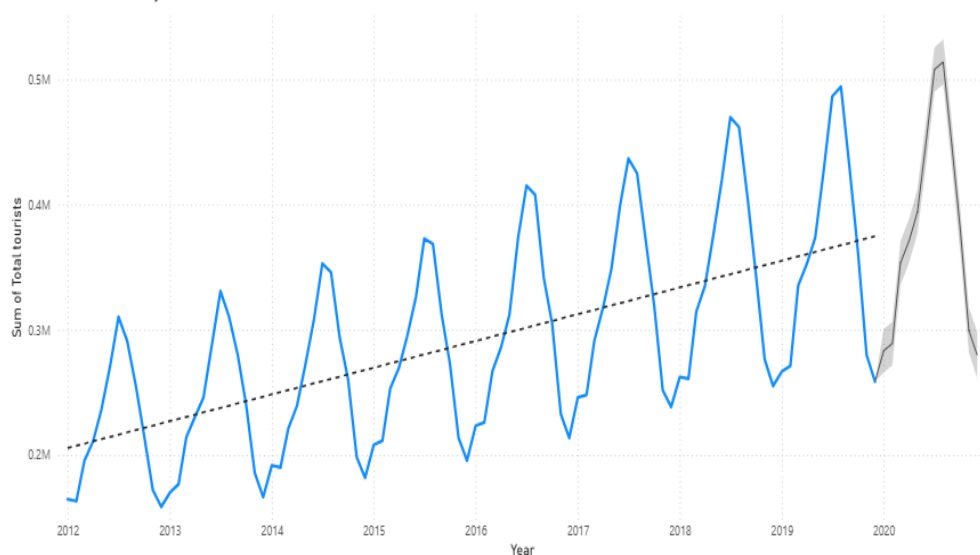
Forecast length	12 points
Ignore last	0 points
Confidence interval	95%
Seasonality	12 points

Then Click Apply

The forecast extends beyond 2019 as a gray shaded area representing the 95% confidence interval.

Step 7: Analyze the Forecast

Sum of Total tourists by Year and Month



Trend

Clear upward trend from ~2.8M (2012) to 4.6M (2019). The model captures consistent year-over-year tourism growth.

Seasonality

Strong peaks in July–August, troughs in Jan–Feb each year. Seasonality = 12 correctly identifies the annual cycle.

Confidence Interval

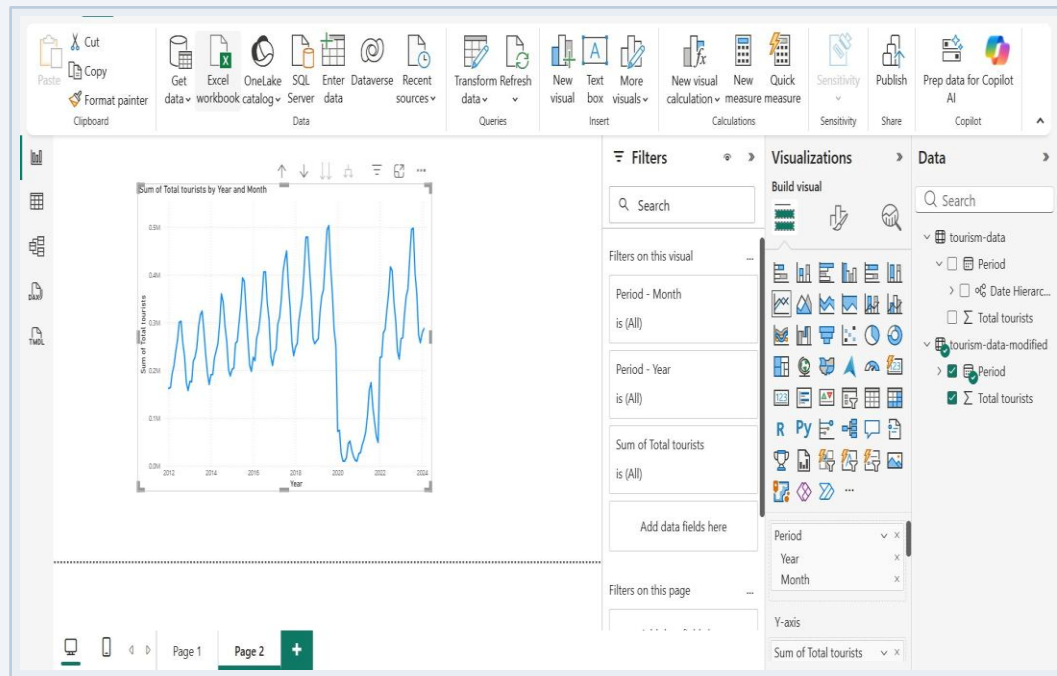
The narrow gray shaded area at 95% CI indicates high model confidence — consistent patterns in training data produce reliable forecasts.

PART 2

Forecasting with Imperfect Data

Dataset: tourism-data-modified.csv • 50 new points with COVID-19 disruption

Part 2 – Imperfect Data Overview



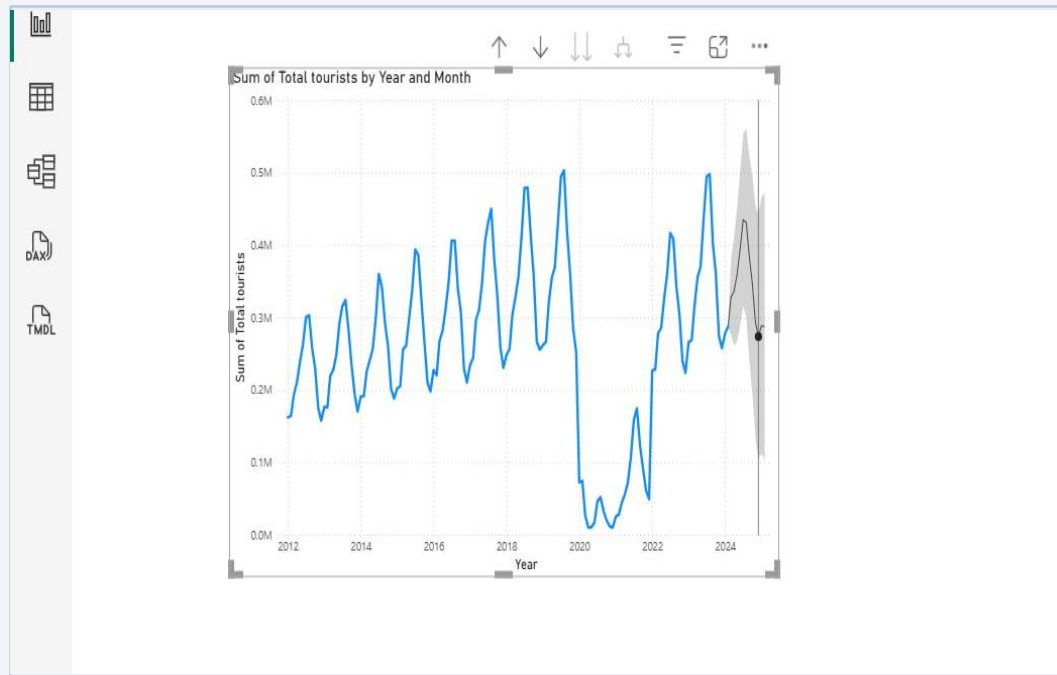
Dataset: tourism-data-modified.csv

- Jan 2012 – Feb 2024 (146 rows)
- 50 new data points beyond original
- COVID disruption simulated in 2020–2021
- Recovery trend through 2022–2024

Phases Observed

2012–2019: Stable growth + seasonal waves
 2020: Near-collapse (lockdowns)
 2021: Partial recovery
 2022–2024: Strong recovery to pre-pandemic levels

Experiment 1: Baseline Forecast (Ignore last = 0, CI = 95%)



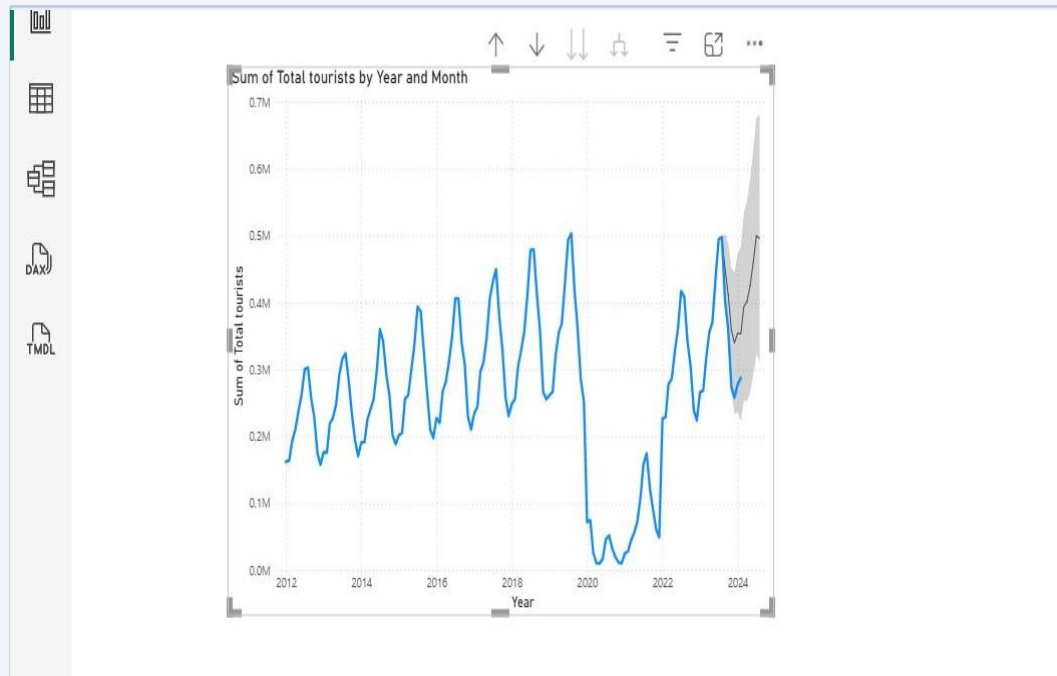
Settings

Forecast length	12
Ignore last	0
Seasonality	12
Confidence interval	95%

Observation

Forecast continues recovery trend. Confidence interval is moderately wide due to COVID disruption in training data. Projects continued seasonal growth.

Experiment 2: Ignore Last 6 Points (CI = 95%)



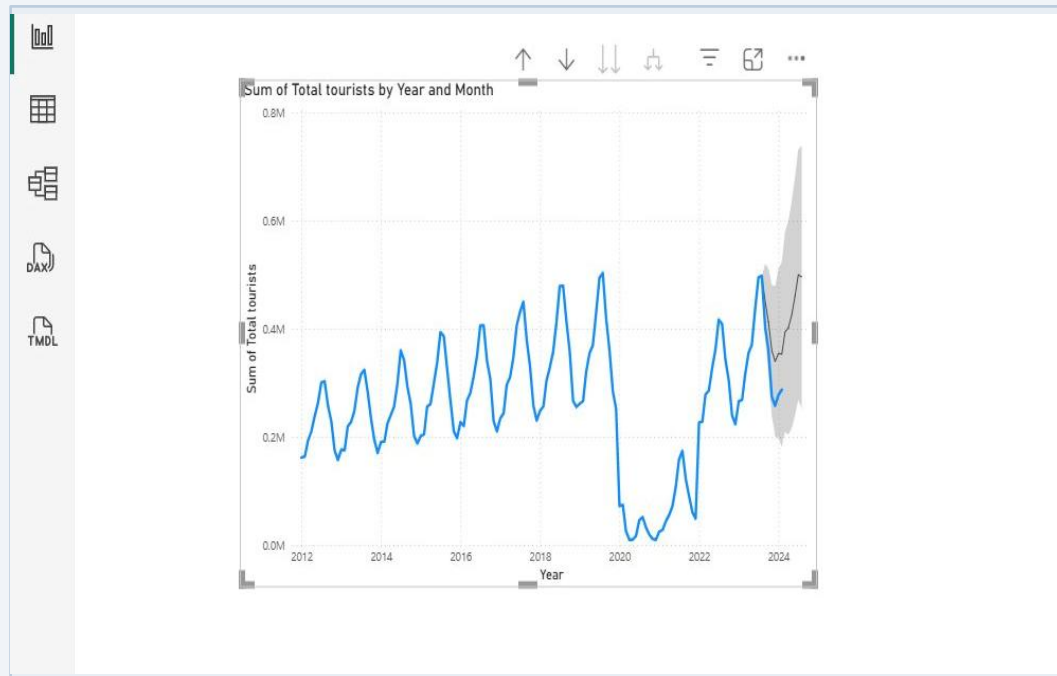
Settings

Forecast length	12
Ignore last	6 ← changed
Seasonality	12
Confidence interval	95%

Observation

Excluding the last 6 months shifts the forecast start earlier. The confidence interval widens slightly — less recent data = more uncertainty in predictions.

Experiment 3: Higher Confidence Interval (CI = 99%)



Settings

Forecast length	12
Ignore last	6
Seasonality	12
Confidence interval	99% ← changed

Observation

The gray shaded area is noticeably wider at 99% CI. To be 99% sure, the model must encompass a much broader range — reflecting high uncertainty from the COVID disruption.

How the Forecast Handled the COVID Disruption

Did NOT Predict the Disruption

Power BI cannot foresee sudden external shocks like COVID-19 from historical time-series data alone. The sudden drop was entirely unpredictable.

Wider Confidence Intervals

All 3 experiments showed wider CI compared to Part 1 — the model accounts for the higher uncertainty introduced by the disruption in training data.

Incorporated Recovery Trend

Post-disruption recovery data (2022–2024) positively influenced the forecast, projecting continued growth following the recovery trajectory.

Seasonal Pattern Disrupted

COVID years broke the regular annual seasonal cycle, reducing the model's confidence in seasonal pattern estimation.

Guide Questions & Answers

What does a narrow confidence interval suggest?

A narrow CI indicates the model has high confidence — the historical data has a consistent and predictable pattern, so the forecast range is tight.

How does Ignore Last affect the forecast?

Setting Ignore Last = 6 excludes the 6 most recent data points from training. The forecast shifts earlier and the CI widens slightly, showing more uncertainty from less recent data.

How does changing CI from 95% to 99% affect the visual?

The shaded gray area becomes noticeably wider. At 99%, the model needs a broader range to guarantee the true value falls within the interval — implying more uncertainty.

How useful is Power BI forecasting for this data?

Highly useful for clean data (Part 1). For imperfect data (Part 2), the wide CI after disruptions limits practical reliability — more advanced modeling may be needed.

Conclusion

✓ Strengths

Easy to use with no coding required. Effective for clean, seasonal data. Configurable settings allow fine-tuning. Confidence intervals communicate reliability visually.

✗ Limitations

Cannot predict external shocks. Wide CI after disruptions reduces reliability. Based on exponential smoothing only — no ARIMA or ML models available.

★ Key Takeaway

Power BI forecasting is ideal for stable time-series. For datasets with major anomalies like COVID-19 disruptions, the forecast should be interpreted with caution and supplemented by domain expertise.